

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

16th March 2017

Subject ST7 – General Insurance: Reserving & Capital Modeling

Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You have then three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet unless instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1) i)** Contrast and evaluate the differences between a claims-made and occurrence policy w.r.t.
- a)** Coverage Trigger (1)
 - b)** Reporting Lag (1)
 - c)** Settlement Lag (1)
 - d)** Sensitivity to inflation (1)
 - e)** Investment income (1)
 - f)** Reserving risk (1)
- ii)** Explain in words or demonstrate diagrammatically that a 1-year Occurrence policy sold on January 1st, 2017 is the same exposure as a first year claims-made policy written on January 1st, 2017 with retroactive date of January 1st, 2017 plus an additional Extended Reporting policy. (1)
- [7]**

Q. 2) You are a consultant for a company which has engaged your services for developing their actuarial reserve estimates. You have been given the following triangles

i) Cumulative Reported Counts

Year	12	24	36	48	60	72	84
2010	180	195	200	215	217	218	218
2011	210	220	225	230	232	233	
2012	212	230	235	240	241		
2013	160	175	185	190			
2014	140	150	160				
2015	138	150					
2016	110						

- a)** Develop an estimate of the ultimate reported counts by year using the chain-ladder method. Justify your selection of appropriate age-to-age factors. (3)
- b)** List the questions you intend to ask the appointed actuary of the company to identify the reasons for the reported patterns above. (2)

In response to your questions, the Appointed Actuary provides the following triangle-

ii) Cumulative Closed Claims

	Lag (in months)						
Year	12	24	36	48	60	72	84
2010	36	49	60	97	130	164	207
2011	42	55	68	104	139	175	
2012	42	58	71	108	145		
2013	64	90	120	173			
2014	56	80	108				
2015	55	78					
2016	44						

- a)** Analyze the claim closure rates and identify possible reasons for the reporting and closing pattern. (3)
- [8]**

Q. 3) i) List and briefly explain the variety of risks faced by a general insurer (3)

ii) Sort the following lines of business in increasing size of actuarial “best estimate” technical reserves and width of 90th percentile reserve ranges. Assume assets, market cycles, underlying book performance etc. are same across the different products. Explain your chosen ordering.

- a) Medical Professional Liability
- b) Car Insurance
- c) Life Insurance
- d) Health Insurance
- e) Credit Card Fraud Insurance

(5)

iii) You are the Appointed Actuary of a general insurance company. The regulatory body for insurance is questioning your approach to reserving by using the data below. The regulator claims that you are consistently “over-reserving” and should instead aim to hold booked reserves at the 50th percentile. The data shows the ultimate incurred loss estimates for an accident year at the end of a given calendar year. All amounts are in millions.

a) Explain why the regulator claims this based on the data below. (1)

b) Provide an appropriate response defending your position as the actuary.

Accident Years	Calendar Years						
	2010	2011	2012	2013	2014	2015	2016
2010	780	760	750	745	740	740	740
2011		810	790	760	750	745	740
2012			797	772	760	750	745
2013				813	784	760	750
2014					739	700	680
2015						750	723
2016							800

(1)

iv) For a particular industry the analysis of the consolidated financial statements of all insurance companies results in the following data. Explain the underlying trends at play in the industry which may give rise to such a phenomenon. All amounts in billion rupees.

Calendar Year	Total Industry Reserve Release
2008	2.1
2009	1.5
2010	0.9
2011	0.6
2012	0.1
2013	-0.2
2014	-0.7
2015	-1.6
2016	-2.1

(2)

[12]

- Q. 4) i)** You are developing the ultimate reserve estimates for a book of business in 2017 and find some older outstanding claims are still at their system generated case reserve levels of INR 2000. In order to develop a more realistic estimate of their ultimate values, you use a summary report available from the national claims data bank which is the industry database for all insurers.

Accident Year	Outstanding Claims	Outstanding Reserves	Type of Claim
2003	1	2,000	Brain Damage
2004	2	4,000	Death
2005	0	-	
2006	1	2,000	Temporary Minor Damage
2007	5	250,000	NA
2008	15	1,125,000	NA
2009	25	2,500,000	NA
2010	37	3,700,000	NA

Industry Data		
Type of Claim	Total Claims	Total Paid Amounts
Brain Damage	345	120,750,000
Death	127	19,050,000
Temporary Minor Damage	532	34,580,000

- a)** Develop an estimate of the yet-to-be-paid amounts for 2003-2006 using industry data (1.5)
- b)** Provide reasons why the use of industry data may not be appropriate in this case (1.5)
- ii)** Compare and contrast the advantages and disadvantages of the Mack method and the Bayesian method for stochastic reserving (4)
- iii)** You are given the following notation –

$U(0)$ = Ultimate claim amounts from prior analysis

$U(CL)$ = Ultimate claim amounts based on chain-ladder method

$U(BF) = U(1)$ = Ultimate claim amounts based on Bornhuetter-Ferguson method

p = Proportion of Ultimate CL claim amounts paid to date. $0 < p < 1$

$q = 1 - p$ = Proportion of unpaid claim amounts. $0 < q < 1$

C = Claim amounts paid to date

- a)** Express $U(BF)$ as a credibility weighted mixture of $U(CL)$ and $U(0)$ (1)

b) Actuary A disagrees with the selection of $U(0)$ and wishes to follow an iterative process for calculating ultimates as described below -

- Step 1 – Apply Bornhuetter-Ferguson method to get $U(1)$
- Step 2 – Repeat step 1 by replacing $U(0)$ with $U(1)$ to get $U(2)$
- Iterate – Repeat step 1 and 2 by replacing $U(n-1)$ with $U(n)$

Express the general ultimate claim amounts $U(n)$ in terms of $U(CL)$ and $U(0)$ (3)

c) Actuary B disagrees with the selection of $U(CL)$ and wishes to follow an iterative process for calculating ultimates as described below -

- Step 1 – Apply Bornhuetter-Ferguson method to get $U(1)$
- Step 2 – Repeat step 1 by replacing $U(CL)$ with $U(1)$ to get $U(2)$
- Iterate – Repeat step 1 and 2 by replacing $U(n-1)$ with $U(n)$

Express the general ultimate claim amounts $U(n)$ in terms of $U(CL)$ and $U(0)$ (3)

d) Show that as n becomes sufficiently large, Actuary A's ultimates become closer to $U(CL)$ and Actuary B's estimates become closer to $U(0)$ (2)

iv) You are developing the reserves for a book of business at the end of 2016 based on data from the previous 9 accident years. Create a chart with the percentile of reserves on the y-axis and accident years grouped into buckets of 3 years on the x-axis. Display the 99th, 75th and 50th percentile reserves for each of the three year buckets. Assume year to year variation in experience is minimal and ignore y-axis units. List any other assumptions you make. Explain the shape of the lines in the chart. (2)

v) Your company is migrating all its legacy claims into a new web-based claims system. As the actuary you are part of a team responsible for testing the data in the legacy and new systems. Your IT team provides the triangle of outstanding reserves (in billions) from the old system

Accident Year	12	24	36	48	60	72	84
2010	1.6	1.2	0.6	0.2	0.1	0	0
2011	1.5	1.4	0.5	0.2	0.1	0	
2012	1.7	1.3	0.6	0.3	0.1		
2013	1.5	1.4	0.5	0.2			
2014	1.6	1.2	0.6				
2015	1.5	1.3					
2016	1.7						

Your finance team provides you with the tests it has run to validate the data migration. It has tested to see if outstanding reserves as of December 31st, 2016 were the same in the old and new systems.

AY	Outstanding Reserves	
	Old System	New System
2010	0	0
2011	0	0
2012	0.1	0.1
2013	0.2	0.2
2014	0.6	0.6
2015	1.3	1.3
2016	1.7	1.7
Total	3.9	3.9

- a) Is this a sufficient test for the data migration initiative? What implications may this have? (1)
- b) Due to time constraints if this is the only test being run, how does your communication of risks to the management differ if your company was in a short-tailed vs. long-tailed line of business (1)
- [20]

- Q. 5) i) Define
- a) Retrocession and the two parties involved in retrocession (1)
- b) Proportional Insurance along with its two common forms (1)
- c) Non-proportional insurance along with its common forms (1)
- ii) Define the following terms with respect to reinsurance
- a) Stability Clause (2)
- b) Rate on line (2)
- iii) Discuss factors you will consider while re-assessing the reinsurance programme of your company. (5)
- [12]

- Q. 6) You are an actuary working for an investment bank. Your company has asked you to screen the General Insurance industry for a potential target company.

Motor is one of the dominant line of business in the industry and Motor Third Party liability reserves has been one of the sources of risk in the industry.

- i) Explain how you would obtain strength of reserves from publically available information. (5)
- ii) Outline the risks and uncertainties faced by Motor Third Party liability business (5)

In your analysis, you have observed few companies with significant Motor Third Party Liability business had also higher proportion of their investment portfolio in equities, compared to other companies.

- iii) Explain the advantages and the disadvantages of such investment strategy (6)
- iv) List factors you would consider while determining the investment strategy of general insurance companies. (5)
- [21]

- Q. 7)** You have been recently hired by a General Insurance company as a Capital Modelling actuary. The company does not have a capital modelling department and it is one of your mandates to establish the capital modeling department. The Board of the company had earlier engaged a consultant to provide a road-map and best practices to be followed by the proposed department. One of the suggestions from the consultant was that the company should go for stochastic modelling
- i)** What are the factors you would have assessed while determining the modelling approach for the company? (3)
 - ii)** What are the advantages and disadvantages of stochastic modelling (7)
 - iii)** What are the various dependencies among variables you would like to build in the stochastic model (5)
 - iv)** The consultant had also emphasized on the role of robust model validations to be put in place
 - a)** What should be the aim of validation (2)
 - b)** Why is validation important (3)
- [20]**
